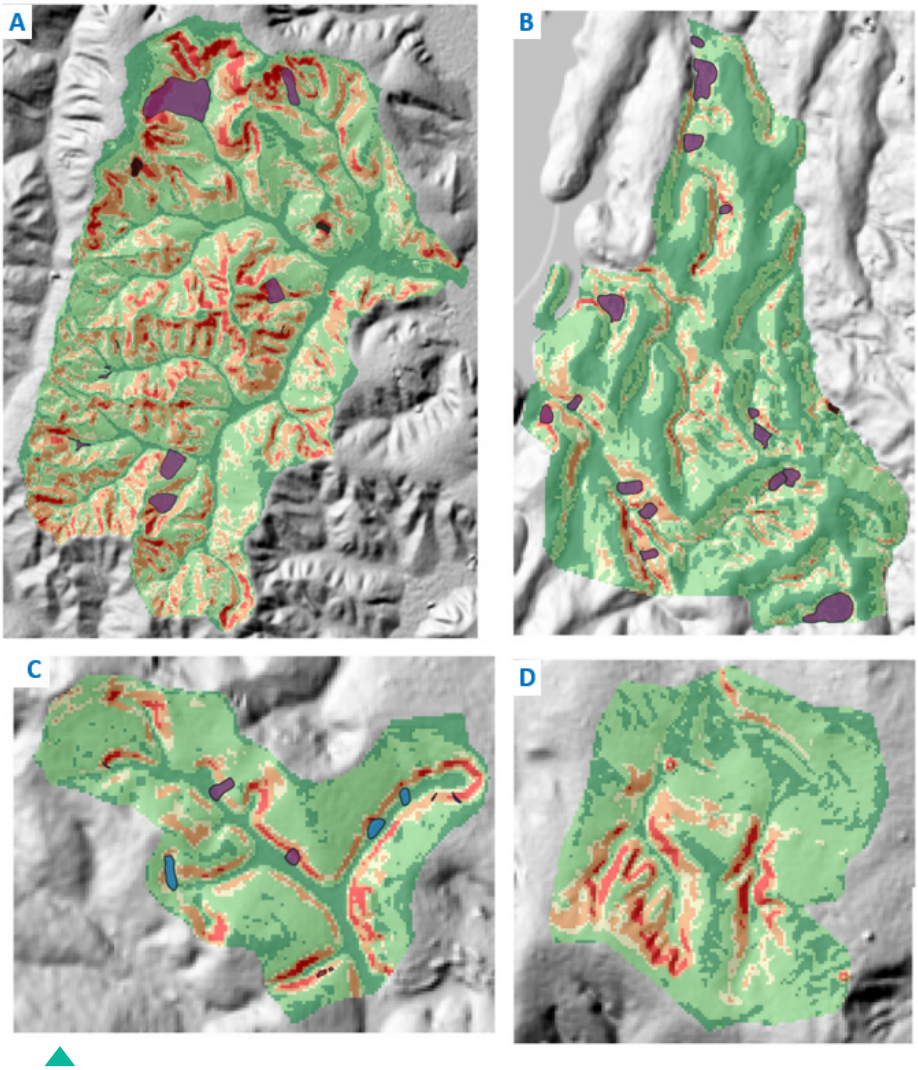
2.4.3.3 Susceptibility maps

The outputs of a logistic regression model, as well as for data-driven models in general, provide values that have no physical meaning. For the relevant interpretation of a data-driven susceptibility model, there is a need for classification (Corominas et al., 2014; Reichenbach et al., 2018). Here, the continuous values of the susceptibility models are classified into five unequally-spaced susceptibility classes (Figure 5). The category [0.80 – 1.0] presents the class that is the most prone to landsliding. The opposite class is ≤ 0.2]. The class]0.45-0.55] present the zone where the uncertainty on the model classification performance is the highest (Rossi et al., 2010). This way of classifying susceptibility models allows comparison between the sub-catchments (e.g. Jacobs et al., 2018).

The susceptibility of the models are presented together with the inventory of the recent and old deep-seated landslides (Figure 5). As explained earlier, these landslides are here considered as a specific susceptibility zonation where the probability of ground deformations is expected to be higher than outside these slope instabilities. We propose two categories (old and recent), the susceptibility to ground deformations being higher for the recent deep-seated landslides.



A - Bishyenyi
B - Rusizi
C - Rwabayanga
D - Rwandex Magerwa

Figure 5: *Landslide susceptibility in the four study areas (purple and blue areas are inventoried landslides)*